

Reading, writing, Recognising

Advances in speech and handwriting recognition promise to move us away from the desktop

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The desktop is at the centre of most computing today. This is the exact opposite of what proponents of Ubiquitous Computing, or Ubicomp, envisage—they believe that computers should become invisible, blending into the fabric of our daily lives. We should not need to know what they're like, or how they work—no more than you need to know about printing technology, how paper is made and where ink comes from. The goal of Ubicomp is silent computers everywhere, not thrusting their presence upon us as today's desktops and laptops do.

With computers all over the place, communication is the imperative. It seems obvious that each machine would require us to learn something new in order to communicate with it—pushing buttons on a telephone, typing on a keyboard, etc. But it's not obvious at all that the machine should dictate how we behave. Intuitive-

ly, it should be the other way round. We should be able to communicate with our machines the way we do with people. Ubicomp is a lofty proposition, but its ideals drive development, albeit not overtly—think PDAs, cell phones and Bluetooth. As a first step towards Ubicomp, we expect that our basic means of communication—speech and writing—should be easily interpreted by our machines.

Today's best speech recognisers approach 99 per cent accuracy, but they often fail when facilitating our interaction with computers. Why? Because the interfaces aren't transparent. Speech recognition (SR) is most visible today in dictation software. The setup involves a headset, which restricts freedom of movement. Next comes the training of the software, and training oneself to speak in such a way that the software understands. Dictation is a skill, not a

natural mode of speech, and worse, small errors can be frustrating; when an error needs correction, you have to switch from speaking to typing. Infusing the software with AI capabilities is not the answer. For a computer to routinely understand natural commands such as "no, not 't', make that a 'd'", it needs to understand natural language—a far harder problem than voice recognition.

We can't yet talk with our machines; we can only instruct them to do specific things. The problem is that we live and talk in 'conversational' mode, within which people understand us. A computer requires us to get into 'command' mode. Experts expect that this will soon change; the state of the art of SR is much more promising than it was just a decade ago. Accuracies have improved, dictation speeds have increased and costs have fallen dramatically.